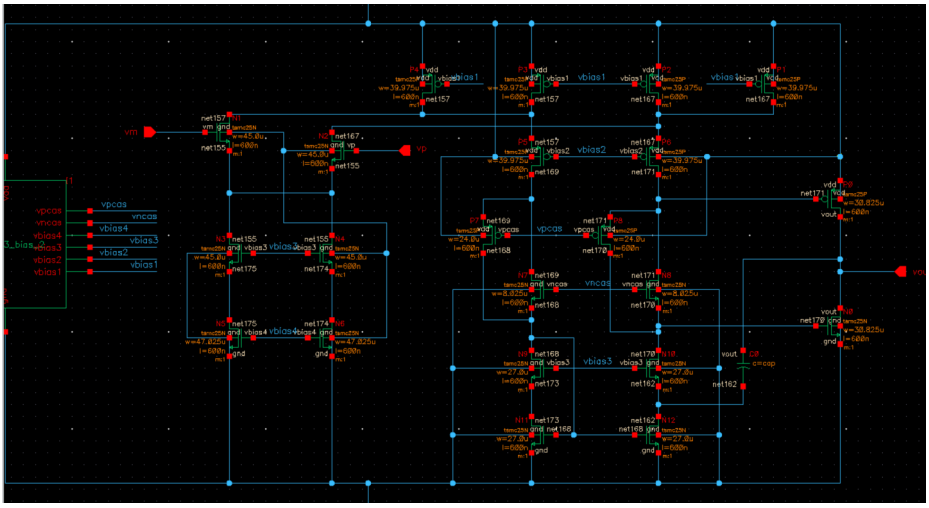
ANALOG IC DESIGN

Folded-cascode CMOS amplifier

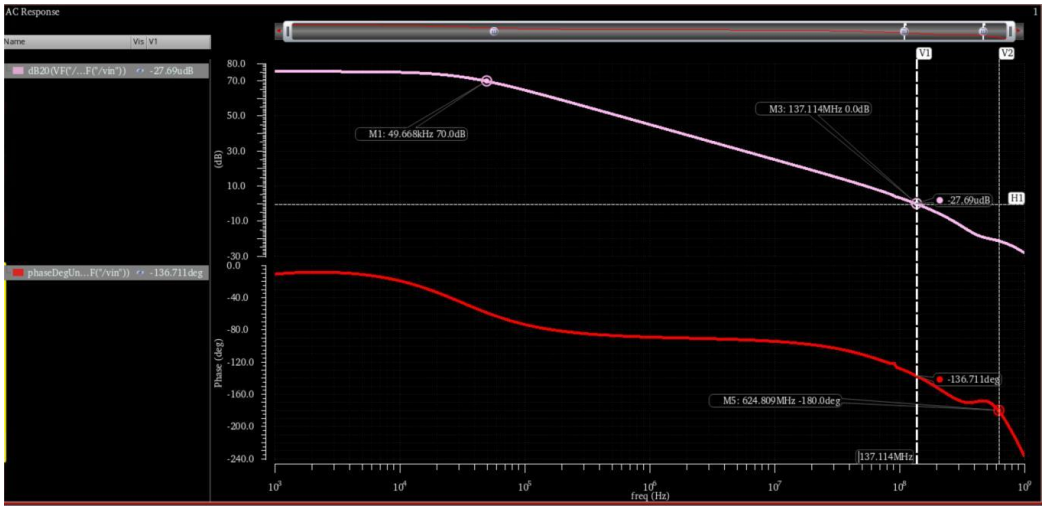
Ruthvik Reddy Anthireddigari / Arizona State University / Spring 2026

A folded-cascode gain stage feeds a Class-AB output buffer. I sized and biased the transistors, then evaluated gain, stability and bandwidth under the specified load conditions.

Transistor-level design



Gain and phase response



The stability simulation gave about 70 dB gain and 57.5° phase margin against a 60° target. A separate 1 pF load simulation gave a unity-gain frequency of 169 MHz.

EEE 523 Advanced Analog IC Design, folded-cascode project report, pp. 4, 9, 15, 21-22.